

MAPQUEST for Formal Proof

A Hilbert–Tenneson Navigation Algorithm for Verifier-Gated Automated Theorem Proving

Graduate Algorithm Specification 001

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Abstract

This paper gives a single explicit algorithmic formulation of a Hilbert-addressed automated theorem prover built from rational formula addresses, tagged rule-arity geometry, finite Hilbert traversals, verifier-gated proof certificates, target-conditioned settlement guidance, and the Tenneson1/Tenneson2/Depth Charge control protocols. The central distinction is that the address of a target formula identifies a destination but does not itself define proof distance. Hilbert curves traverse candidate *inference applications*, not formulas. A proof is therefore a verifier-accepted route through a directed hypergraph whose vertices are zero-dimensional well-formed formulas and whose hyperedges are legal rule applications. For a rule of arity k , ordered premise slots induce an intrinsic k -dimensional rational action component; schema instantiations, substitutions, and side conditions form an effectively enumerable fiber over that component. Finite Hilbert bijections organize the search of each rule component, while a separate settlement-distance estimator, structural target compass, Professor partial credit, creativity, Dreamer proposals, four oracle interfaces, Depth Charge escape, Sentinel resource control, and a mandatory fair fallback determine how the search budget is allocated. The resulting system is presented as an algorithm rather than as a geometric analogy. Its trusted boundary remains small: search mechanisms may alter ordering and resource allocation, but only the verifier can admit mathematical statements to the verified BANK.

Central principle.

The target address identifies the destination;
Hilbert traversal orders candidate inference roads.

The curve does not prove the theorem, and numerical proximity of two formula addresses is not automatically proof proximity. Target-directed navigation requires a separate proof-potential or settlement-distance layer.

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1 Purpose and algorithmic viewpoint

The goal is to specify one recognizably algorithmic object, in the same methodological spirit in which one can refer to Buchberger’s algorithm rather than to an informal collection of polynomial-reduction ideas. The analogy is limited: the present search problem is generally semi-decidable rather than a terminating decision problem, and its action space may be enormous. Nevertheless, one can still give explicit state variables, legal actions, ordering rules, update equations, invariants, stopping conditions, and certificate semantics.

The proposed name is *Hilbert–Tenneson MAPQUEST*. The word “MAPQUEST” is not used to claim that theorem search has become ordinary shortest-path routing. It names a decomposition:

$$\text{addressing} \rightarrow \text{road atlas} \rightarrow \text{navigation policy} \rightarrow \text{verified route}.$$

The exact route is a formal proof certificate. The map is the enumerable space of legal inference applications. The navigator controls search order. The verifier remains the sole proof authority.

2 Formal environment and trusted boundary

Definition 2.1 (Frozen proof environment). *A proof run is parameterized by a tuple*

$$\mathfrak{E} = (\Sigma, \mathcal{W}, \Gamma, \mathcal{R}, \text{Side}, \mathbf{V}, \nu, T),$$

where Σ is the signature, $\mathcal{W}$ the effectively enumerable WFF domain, $\Gamma \subseteq \mathcal{W}$ the admitted axioms and hypotheses, $\mathcal{R}$ a finite or effectively enumerable set of finitary rules, Side the decidable or semi-decidable side-condition machinery required by those rules, $\mathbf{V}$ the certificate verifier, ν the frozen environment/version identifier, and $T \in \mathcal{W}$ the target.

The verifier is treated as a trusted kernel relative to the declared environment. Search modules may be heuristic, learned, stochastic, reflective, or oracle-assisted. None may create theoremhood by fiat.

Principle 2.2 (Verifier sovereignty). *For every mathematical candidate C and certificate fragment π ,*

$$C \in \text{BANK}_{t+1} \setminus \text{BANK}_t \implies \mathbf{V}(\mathfrak{E}, C, \pi) = 1.$$

No Professor score, Dreamer proposal, oracle output, Hilbert location, or controller confidence can substitute for this test.

3 Formula identity and Tenneson rational addressing

3.1 Zero-dimensional WFFs

Definition 3.1 (Intrinsic formula dimension). *Each WFF $A \in \mathcal{W}$ is a vertex and has intrinsic dimension*

$$\dim(A) = 0.$$

Any rational or natural number attached to A is address metadata, not a geometric dimension of the formula itself.

Let

$$g : \mathcal{W} \hookrightarrow \mathbb{N}_0$$

be a computable injective formula line numbering. In a concrete formal library this may be a canonical syntax code, a WFF line, or another frozen injective code.

Definition 3.2 (Injective rational formula address). *Define*

$$a(A) = \frac{g(A)}{g(A) + 1} \in \mathbb{Q} \cap [0, 1).$$

Then a is injective and preserves exact formula identity through g .

3.2 The explicit natural-to-rational layer

The project also uses an explicit surjection

$$r : \mathbb{N}_+ \twoheadrightarrow \mathbb{Q}$$

from Tenneson's natural-to-rational construction, together with a least-preimage section s satisfying

$$r(s(q)) = q$$

for reduced $q \in \mathbb{Q}$.

Transport this enumeration to the rational unit interval by

$$u(1) = 1, \quad u(n) = \frac{|r(n-1)|}{1 + |r(n-1)|}, \quad n \geq 2.$$

Then

$$u : \mathbb{N}_+ \twoheadrightarrow \mathbb{Q} \cap [0, 1].$$

For $0 \leq q < 1$, one may use

$$u_{\min}^{-1}(q) = s\left(\frac{q}{1-q}\right) + 1, \quad u_{\min}^{-1}(1) = 1.$$

Definition 3.3 (Canonical whole-number formula address). *For $A \in \mathcal{W}$, define*

$$n(A) = \nu_{\min}^{-1}(a(A)).$$

The triple

$$\tau(A) = (g(A), a(A), n(A))$$

is the exact destination record used by MAPQUEST.

Remark 3.4 (Why the injective layer matters). The surjection ν alone cannot identify a formula because a surjection may map multiple naturals to the same rational. Formula identity is therefore

carried by $g(A)$ (or equivalently by the injective $a(A)$), while the least-preimage machinery supplies a canonical natural-number coordinate for navigation and provenance.

3.3 Tuple addresses

Use the pairing polynomial

$$\pi(a, b) = \frac{(a + b - 1)(a + b - 2)}{2} + a$$

and recursively obtain a bijection

$$\pi_k : \mathbb{N}_+^k \rightarrow \mathbb{N}_+.$$

For an ordered premise tuple $(A_1, \dots, A_k)$ define

$$N(A_1, \dots, A_k) = \pi_k(n(A_1), \dots, n(A_k)).$$

Thus every ordered premise tuple has both a rational coordinate vector and a canonical whole-number address.

4 Rules define the roads: tagged rule-arity geometry

Definition 4.1 (Rule component). *Let $r \in \mathcal{R}$ have arity $k = \text{ar}(r)$. Its intrinsic premise-address component is*

$$X_r = \{r\} \times D^k, \quad D = \mathbb{Q} \cap [0, 1].$$

For $k = 0$, take $D^0 = \{\star\}$.

Definition 4.2 (Tagged action coproduct). *For a rule family $z \subseteq \mathcal{R}$, the intrinsic heterogeneous action space is*

$$X_z = \bigsqcup_{r \in z} \left(\{r\} \times D^{\text{ar}(r)} \right).$$

The rule tag is part of the address.

Proposition 4.3 (Local dimension). *A candidate application of a k -ary rule has intrinsic premise geometry of dimension k , independent of the number or arities of other rules in the formal system.*

Proof. The component containing the application is $\{r\} \times D^k$. The discrete tag $\{r\}$ contributes no continuous coordinate, while the k ordered premise slots contribute the k rational coordinates. The coproduct does not identify unrelated components. $\square$

For premises $A_1, \dots, A_k$, the intrinsic inference address is

$$\alpha(r; A_1, \dots, A_k) = (r, a(A_1), \dots, a(A_k)).$$

A whole-number variant is

$$\hat{\alpha}(r; A_1, \dots, A_k) = (r, N(A_1, \dots, A_k)).$$

5 The missing production detail: witness fibers

Premise arity is not necessarily the entire operational action space in a schema-based prover. A rule application may also require a substitution, metavariable assignment, unifier, distinct-variable witness, type witness, or other side-condition evidence.

Definition 5.1 (Instantiation fiber). *For a rule r and ordered premise tuple $\vec{A} = (A_1, \dots, A_k)$, let*

$$\Xi_r(\vec{A})$$

be the effectively enumerable set of admissible instantiation witnesses for that tuple. A complete candidate action is

$$c = (r; \vec{A}; \xi), \quad \xi \in \Xi_r(\vec{A}).$$

Definition 5.2 (Action bundle). *Over a finite verified set $B \subseteq \mathcal{W}$, the operational rule bundle is*

$$\mathfrak{A}_r(B) = \bigsqcup_{\vec{A} \in B^k} \Xi_r(\vec{A}).$$

The base B^k carries the intrinsic k -dimensional Hilbert geometry; $\Xi_r(\vec{A})$ is an enumerable fiber over each base point.

This base/fiber distinction preserves the rule-arity geometry while making the algorithm adequate for proof systems in which premises alone do not determine a unique rule instance.

6 Finite Hilbert traversal of a frozen verified set

Let the verified BANK at epoch e be frozen as the ordered list

$$B_e = (\theta_0, \dots, \theta_{M-1}).$$

Choose

$$p = \lceil \log_2 M \rceil, \quad N = 2^p.$$

Definition 6.1 (Finite Hilbert approximant). *For dimension $k \geq 1$, resolution p , and declared orientation o , a finite Hilbert approximant is a bijection*

$$H_{k,p,o} : \{0, \dots, N^k - 1\} \longrightarrow \{0, \dots, N - 1\}^k.$$

For $k = 0$, use the unique map on the singleton grid.

At Hilbert tick h ,

$$H_{k,p,o}(h) = (j_1, \dots, j_k).$$

If each $j_i < M$, decode the ordered premise tuple

$$(\theta_{j_1}, \dots, \theta_{j_k}).$$

Coordinates with some $j_i \geq M$ are padding and generate no premise application.

Theorem 6.2 (Finite component coverage). *For a frozen B_e of size M and a k -ary rule, a completed traversal of $H_{k,p,o}$ visits every ordered tuple in B_e^k exactly once, ignoring padded cells.*

Proof. Each valid tuple corresponds to a unique coordinate vector in $\{0, \dots, M-1\}^k$, which is a subset of the padded grid. Bijectivity of $H_{k,p,o}$ visits each grid coordinate exactly once. Therefore each valid tuple is decoded exactly once. $\square$

Caution 6.3 (Hilbert order does not reduce exhaustive cardinality). For a completed frozen component there are still exactly M^k valid ordered premise tuples. Hilbert geometry changes visitation order and locality properties; it does not magically reduce the finite candidate count. Any speed advantage must appear in interrupted prefixes, locality, cache behavior, target-conditioned allocation, reuse, or another downstream mechanism.

7 The probe

Definition 7.1 (Complete MAPQUEST search state). *At time t , let*

$$x_t = (T, \mathfrak{E}, B_t, G_t, F_t, C_t, \Sigma_t, R_t, L_t),$$

where B_t is the verified BANK, G_t the verified derivation DAG, F_t the active frontier, C_t the collection of rule/Hilbert cursors, Σ_t certified controller/self-state, R_t resource state, and L_t append-only provenance log.

Definition 7.2 (Probe state). *A probe is*

$$P_t = (x_t, r_t, b_t, m_t, o_t, h_t, \xi_t),$$

where r_t is the selected rule component, b_t a coarse structural/bin descriptor if used, m_t resolution, o_t Hilbert orientation, h_t Hilbert index, and ξ_t the current witness-fiber cursor.

The probe therefore remembers enough state to resume after interruption. It is not merely “the point currently occupied on a Hilbert curve.”

8 Why the target address is not a proof-distance coordinate

The target address

$$\tau(T) = (g(T), a(T), n(T))$$

identifies the exact formula that terminates a successful proof. It does not follow that

$$|a(A) - a(T)|$$

measures proof difficulty or derivational distance.

Principle 8.1 (Address-distance separation). *MAPQUEST maintains three conceptually distinct objects:*

$$\text{formula identity address} \neq \text{inference navigation coordinate} \neq \text{settlement distance}.$$

The first identifies a WFF, the second orders candidate rule applications, and the third estimates how much verified work remains before settlement.

This separation prevents a category error: a syntactic numbering can be an excellent identifier and a poor geometric metric at the same time.

9 Target-conditioned structural navigation

Let

$$\sigma(A) \in \mathbb{R}^d$$

be a theoremhood-independent structural signature of a formula. Possible coordinates include symbol count, syntax-tree node count, formula depth, term depth, quantifier rank, constructor profile, arity profile, repeated-subtree features, recursion profile, type features, and other syntactically decidable invariants.

Definition 9.1 (Structural target distance). *For a declared scaling matrix $W_\sigma \succeq 0$, define*

$$d_\sigma(A, T) = \|W_\sigma(\sigma(A) - \sigma(T))\|.$$

This is a search heuristic and carries no theoremhood semantics.

The signature and its normalization must be frozen before benchmark evaluation to avoid theoremhood leakage.

10 Shortest settlement and the ideal navigator

Let $\mathcal{A}(x)$ denote the legal control actions available from complete state x . An action may select an exact inference candidate, refine a coarse Hilbert cell, invoke a bounded controller mode, allocate compute to an agent, or execute another verifier-safe transition. Let

$$F_c(x)$$

be the successor state produced by legal action c .

Let $\mathcal{S}$ be the terminal set of verifier-accepted settlements. In a proof-only run, $\mathcal{S}$ contains states with a verified proof of T . In a broader P/R/I/C architecture it may contain proof, refutation, certified independence, or inconsistency states under the declared certificate semantics.

Definition 10.1 (Optimal settlement distance). *In the unit-cost model define*

$$r^*(x) = \min\{m : \text{a legal length-}m \text{ action sequence takes } x \text{ to } \mathcal{S}\},$$

with $r^(x) = +\infty$ when no terminal state is reachable.*

Theorem 10.2 (Bellman equation). *On the finite-settling basin, for nonterminal x ,*

$$r^*(x) = 1 + \min_{c \in \mathcal{A}(x)} r^*(F_c(x)).$$

An action is shortest-settlement optimal exactly when its successor has value $r^(x) - 1$.*

Proof. Every settling trajectory has a first action c followed by a settling suffix from $F_c(x)$. The shortest trajectory therefore has length one plus the minimum shortest suffix length. Conversely, concatenating an action attaining the minimum with a shortest suffix yields a shortest trajectory from x . $\square$

Thus the ideal MAPQUEST navigator would choose

$$c^*(x) \in \arg \min_{c \in \mathcal{A}(x)} r^*(F_c(x)).$$

The difficulty is computational: exact r^* is generally unavailable on hard theorem-search classes. MAPQUEST therefore uses an estimator.

11 Settlement-horizon estimation and regret

Let

$$\hat{r}_t(x)$$

be the current estimated remaining settlement cost. For an action c define the estimated one-step value

$$\hat{Q}_t(c) = c_t(c) + \hat{r}_t(F_c(x_t)),$$

where $c_t(c) \geq 0$ is the declared immediate action cost.

Definition 11.1 (Estimated settlement-horizon regret).

$$\text{Reg}_t(c) = \hat{Q}_t(c) - \min_{b \in \mathcal{A}_t} \hat{Q}_t(b).$$

A candidate with zero estimated regret is tied for the best currently predicted continuation.

This is an online surrogate. Exact certificate-horizon regret can be evaluated retrospectively when a canonical minimum-cost certificate is available; discovery regret requires a separately declared search-cost model.

12 Professor partial credit and throttling

Definition 12.1 (Professor score). *For a candidate action c , the Professor returns*

$$P_t(c) \in [0, 1],$$

a heuristic assessment of partial progress toward the declared target or settlement objective.

The Professor is advisory. It cannot verify mathematics.

Let

$$\lambda_t \in [0, \lambda_{\max}]$$

be a dynamic throttle. The effective Professor contribution is

$$P_t^{\text{eff}}(c) = \lambda_t P_t(c).$$

A controller may increase λ_t when held-out calibration is good and reduce it under stagnation, herding, instability, or distribution shift. In torpor, λ_t may approach zero while the base fair search continues.

13 The MAPQUEST routing functional

A practical target-conditioned action score can combine settlement estimates, structural proximity, Hilbert locality, immediate resource cost, reusable BANK value, and Professor partial credit.

Definition 13.1 (Routing functional). *For candidate action c at time t , define*

$$J_t(c; T) = w_R \text{Reg}_t(c) + w_S d_\sigma(C(c), T) + w_H d_H(c, \mathcal{G}_t) + w_C \text{cost}_t(c) - w_B \text{BankValue}_t(c) - \lambda_t P_t(c).$$

Here $C(c)$ is the syntactic candidate conclusion when defined, d_H is a declared local Hilbert travel/refinement cost, and $\mathcal{G}_t$ is the currently active geometric region.

The coefficients are engineering parameters, not logical constants. They must be frozen or tuned only on permitted development data.

Principle 13.2 (Fast policy). *The guided navigator approximately selects*

$$c_t \in \arg \min_{c \in \mathcal{A}_t} J_t(c; T),$$

subject to Sentinel admissibility and the mandatory fair-reserve schedule defined below.

14 Multiresolution MAPQUEST

Enumerating billions of exact Hilbert cells merely to score them defeats the point of a navigation layer. The search should operate on hierarchical Hilbert intervals.

At resolution m , let

$$I_{r,m,j}$$

denote a contiguous Hilbert interval corresponding to a coarse dyadic region of the r -component. Define a cheap lower-bound or optimistic priority estimate

$$L_t(I) \approx \inf_{c \in I} J_t(c; T).$$

The coarse-to-fine procedure is:

$$\text{component} \rightarrow \text{coarse cell} \rightarrow \text{subcell} \rightarrow \cdots \rightarrow \text{exact candidate}.$$

Remark 14.1 (What can make this faster). Hilbert order does not reduce the final number of exhaustive candidates. Speed can arise only if the representation has useful locality and the algorithm is interrupted before exhaustive completion, or if local ordering improves cache behavior, lemma reuse, verifier batching, branch reuse, or target-mass concentration.

15 Tenneson1: the navigation protocol

Tenneson1 (T1P) is treated as the primary navigation policy:

$$\text{T1P} = \text{Hilbert multiresolution navigation} + \text{Creativity} + \text{Minimum Settlement-Horizon Regret}.$$

The Hilbert component provides systematic local traversal. The creativity component injects bounded alternative proposals outside the current best-first neighborhood. The horizon component favors actions predicted to reduce remaining settlement cost.

15.1 Creativity reserve

Let $\varepsilon_t \in [0, 1]$ be a declared creativity/fairness mixture weight. At time t the controller may choose a creative action with probability or scheduled frequency ε_t , provided all mathematical outputs remain verifier-gated. For eventual fairness under a stochastic mixture, a sufficient condition is that the background fair channel receive infinitely many turns; for example,

$$\sum_{t=1}^{\infty} \varepsilon_t = \infty$$

for the fair component when independence assumptions appropriate to the scheduler are satisfied.

16 Tenneson2: syntax-first execution and reproducibility

Tenneson2 (T2P) is the execution envelope around T1P. Before search begins, freeze and record at least:

(formal database hash, T , V , certificate language, rule set,
address map, Hilbert family, budgets, seed, allowed lemmas).

The target proof, when benchmarking, must be masked from the search process. A stored proof may not leak through training features, retrieval, cached labels, or preprocessing artifacts.

Principle 16.1 (T2P semantics). *A run may return a verified settlement certificate or UNRESOLVED under exhausted declared resources. Timeout or search failure is not refutation and not an independence certificate.*

17 Depth Charge as bounded nonlocal escape

Let $\text{Stag}(x_t)$ be a declared stagnation predicate based on quantities such as lack of BANK growth, poor settlement-horizon progress, repeated local failure, or frontier degeneracy.

When stagnation fires, save

$$x_{\text{save}} = x_t$$

and allocate a fixed budget B_{DC} . A Depth Charge may alter orientation, region, rule allocation, search direction, representation, backward/forward balance, induction/rewrite emphasis, or agent allocation. It may not alter verifier semantics.

Principle 17.1 (Depth Charge rollback). *At the end of a Depth Charge, restore the saved search state except for independently verified deposits and declared controller statistics. Unverified speculative products are discarded or quarantined outside BANK.*

Depth Charge is therefore a bounded escape operator from a locally unproductive navigation basin.

18 Dreamer

Let $\mathcal{H}_t$ be a bounded history containing verified deposits, failed branches, partial-credit signals, resource traces, representation choices, and previous strategy outcomes.

Definition 18.1 (Dreamer). *A bounded Dreamer episode is a map*

$$\mathcal{D}_t : \mathcal{H}_t \times B_D \longrightarrow \mathcal{Q}_t,$$

where B_D is the dream budget and $\mathcal{Q}_t$ is a set of untrusted proposals such as candidate lemmas, branch recombinations, alternative representations, strategy motifs, or heuristic changes.

Dreamer operates on accumulated experience rather than directly on the trusted theorem set. Its products may enter FUTUREBANK or an untrusted proposal queue, but

$$\mathcal{D} \not\rightarrow \text{BANK}.$$

Any mathematical proposal must pass ordinary verification.

19 The four oracle interfaces

The architecture distinguishes four idealized information interfaces:

$$O_1 = \text{role oracle}, \quad O_2 = \text{resource oracle}, \quad O_3 = \text{strategy oracle}, \quad O_4 = \text{certificate oracle}.$$

Their MAPQUEST interpretations are:

O_1	Which settlement lane should dominate: proof, refutation, independence, contradiction/integrity, or unresolved?
O_2	Where should the next unit of search or dream budget be allocated?
O_3	Which action class, representation move, Hilbert region, or search strategy should be tried next?
O_4	Supply a purported terminal certificate candidate.

Even for O_4 ,

$$O_4(T) = \pi$$

means only that π is proposed. Deposit requires

$$V(\mathfrak{C}, T, \pi) = 1.$$

The oracle layer can therefore be used experimentally without enlarging the trusted proof kernel.

20 Sentinel

Sentinel is orthogonal to theoremhood. Let a resource-policy vector be

$$R_{\max} = (\text{RAM, wall time, branch count, replay budget, verifier queue, } \dots).$$

Before expensive action execution,

$$\text{Sentinel}(x_t, c) \in \{\text{ALLOW, THROTTLE, VETO}\}.$$

Sentinel may block resource-pathological search even when the proposed mathematical direction is promising. Conversely, Sentinel may not accept an invalid certificate.

21 Fair reserve and the completeness backstop

Fast navigation must not permanently starve the exhaustive route.

Definition 21.1 (Fair reserve). *Let $\text{FairNext}(x_t)$ be the next action prescribed by a fixed dovetailed exhaustive enumeration of rule components, premise tuples, witness fibers, and resolutions. Choose an infinite schedule $\mathcal{T}_{\text{fair}} \subseteq \mathbb{N}$ such that every finite exhaustive action is eventually reached by the fair channel if the run continues indefinitely.*

A simple deterministic policy is

$$t \equiv 0 \pmod{K} \implies c_t = \text{FairNext}(x_t)$$

for fixed $K \geq 1$.

Theorem 21.2 (Fair-acceleration preservation). *Assume the underlying proof calculus has effectively enumerable finite certificates, the verifier accepts exactly valid certificate steps, and the fair channel*

eventually executes every finite prefix required by any valid finite proof. Interleaving arbitrary additional Hilbert, learned, creative, Professor, Dreamer, or oracle-guided actions cannot destroy eventual reachability of that proof, provided the fair channel is never permanently disabled.

Proof. Let π be a finite valid proof. By hypothesis, each finite action required to reconstruct π appears after finitely many turns of the fair channel. Additional guided actions insert delays but do not delete fair turns. Hence every required finite prefix is eventually executed, after which the verifier accepts the final certificate. $\square$

22 The complete Hilbert–Tenneson MAPQUEST algorithm

Algorithmic specification

Input: frozen environment $\mathfrak{E}$, target T , budgets, Hilbert family, address version, T1P/T2P controller settings, Sentinel policy.

Output: verifier-accepted proof/settlement certificate, or UNRESOLVED on budget exhaustion.

- M1. Freeze and address.** Hash the environment; seal T ; compute $\tau(T)$; initialize $B_0 = \Gamma$, the verified derivation DAG, logs, resource state, and all protocol versions.
- M2. Construct the road atlas.** For each $r \in \mathcal{R}$, construct its tagged intrinsic component X_r , finite Hilbert indexing family, and effectively enumerable witness fiber Ξ_r .
- M3. Initialize coarse regions.** Represent unexplored action space by coarse Hilbert intervals rather than immediately materializing all exact candidates.
- M4. Score regions.** Compute cheap target-conditioned priorities $L_t(I)$ from structural signatures, estimated settlement horizon, local productivity, rule compatibility, Professor guidance, BANK reuse value, and resource cost.
- M5. Choose guided or fair mode.** If $t \in \mathcal{T}_{\text{fair}}$, choose $\text{FairNext}(x_t)$. Otherwise T1P chooses the currently preferred region/action, with bounded creativity.
- M6. Refine.** If a selected region is coarse, replace it with Hilbert subregions and rescore until an exact candidate action $c = (r; \vec{A}; \xi)$ is obtained.
- M7. Construct candidate syntax.** Apply the declared rule implementation to obtain a candidate conclusion $C(c)$ when side conditions and instantiation permit. If $C(c) = T$, give the candidate terminal priority but do not declare success.
- M8. Apply target control.** Evaluate $\hat{Q}_t(c)$, $\text{Reg}_t(c)$, $P_t(c)$, structural distance, BANK value, and the routing functional $J_t(c; T)$ as needed by the active controller.
- M9. Pass Sentinel.** Execute only if Sentinel allows the action within the declared resource policy. Throttled actions may be deferred or reduced in budget.
- M10. Verify.** Send the proposed inference/certificate fragment to V. Rejected candidates are logged as failed roads. Accepted candidates extend the derivation DAG and are deposited in BANK with complete provenance.

- M11. Update the map.** New verified formulas create new possible premise tuples. Update relevant frozen-epoch queues, structural indices, region scores, and BANK-value estimates according to the declared epoch policy.
- M12. Check target.** If the verifier accepts a certificate whose final conclusion is exactly T , reconstruct and emit the Hilbert-addressed proof route and halt successfully.
- M13. Handle stagnation.** If the declared stagnation predicate fires, invoke a bounded Depth Charge. Dreamer may separately generate untrusted counterfactual proposals. Oracle interfaces may intervene only in experimental arms explicitly declaring them.
- M14. Continue or stop.** If resources remain, repeat. If the declared budget is exhausted without a verified settlement, return UNRESOLVED and the interruption certificate.

23 Compact pseudocode

Listing 1: Hilbert–Tenneson MAPQUEST, abstract executable form

```

1  INPUT E, target T, budgets B, protocol settings P
2
3  freeze_and_hash(E, T, P)
4  BANK := verified_seed(E.Gamma)
5  DAG := seed_dag(BANK)
6  ATLAS := build_tagged_rule_components(E.rules, BANK)
7  FAIR := initialize_exhaustive_dovetail(E, BANK)
8
9  while resources_remain():
10     x := complete_state(T, E, BANK, DAG, ATLAS, controllers, resources)
11
12     if fair_turn():
13         c := FAIR.next(x)
14     else:
15         I := best_coarse_region_by_L(x, T)
16         while not exact_action(I):
17             I := refine_best_hilbert_subregion(I, x, T)
18         c := decode_action_with_witness_fiber(I)
19         c := T1P_adjust(c, x, T) # horizon regret + creativity
20
21     if Sentinel(x, c) == VETO:
22         log_veto(c)
23         continue
24
25     candidate := apply_rule_schema(c)
26     if candidate is undefined:
27         log_illegal(c)
28         continue
29
30     score := routing_score(candidate, c, x, T, Professor)
31     log_proposal(c, candidate, score)
32
33     receipt := Verifier(E, candidate, c.certificate_fragment)
34     if receipt.accepted:
35         BANK.deposit(candidate, receipt, provenance=c.address)
36         DAG.extend(candidate, c, receipt)
37         update_atlas_after_verified_deposit(ATLAS, candidate)
38
39         if candidate == T and Verifier.accepts_full_target_certificate(DAG, T):
40             return VERIFIED_PROOF(reconstruct_route(DAG, T))
41

```



```

42     if stagnation_triggered(x):
43         depth_charge_from_checkpoint(x)
44
45     dreamer_may_propose_untrusted_candidates()
46
47     return UNRESOLVED(interruption_certificate())
    
```

24 The proof certificate as a MAPQUEST route

Suppose the target proof contains m verified rule applications. A Hilbert-addressed certificate may be represented schematically as

$$\Pi_T = \begin{bmatrix} (r_1; \alpha_{11}, \dots, \alpha_{1k_1}; \xi_1; \tau(C_1)) \\ (r_2; \alpha_{21}, \dots, \alpha_{2k_2}; \xi_2; \tau(C_2)) \\ \vdots \\ (r_m; \alpha_{m1}, \dots, \alpha_{mk_m}; \xi_m; \tau(T)) \end{bmatrix}.$$

Each α_{ij} names an actual premise address, each ξ_i records the instantiation/side-condition witness needed by the rule, and each conclusion address is checked against the canonical WFF identity map.

The proof is therefore a directed hypergraph route

$$\text{BANK}_0 \rightsquigarrow \alpha_1 \rightsquigarrow C_1 \rightsquigarrow \alpha_2 \rightsquigarrow \dots \rightsquigarrow T,$$

not a continuous path from $a(A)$ to $a(T)$ in a one-dimensional formula line.

25 A schematic architecture

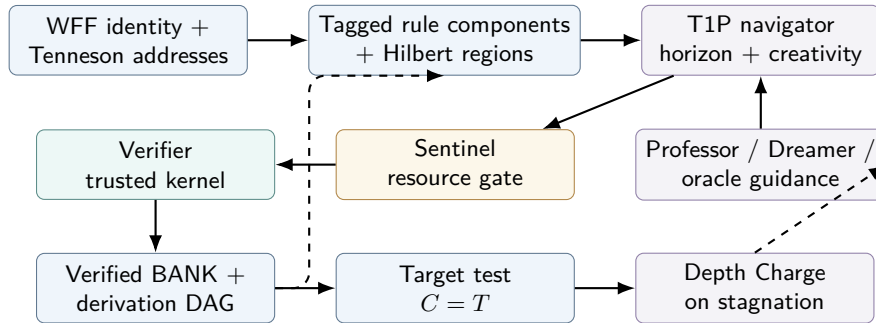


Figure 1: MAPQUEST separates addressing, search geometry, control, resource safety, and proof authority. Only the verifier can create trusted mathematical deposits.

26 Core optimization formula

A compact mathematical control law for the guided part of MAPQUEST is

$$c_t \in \arg \min_{\substack{c \in \mathcal{A}(x_t) \\ \text{Sentinel}(x_t, c) \neq \text{VETO}}} [\text{cost}_t(c) + \hat{r}_t(F_c(x_t)) + \beta d_\sigma(C(c), T) - \lambda_t P_t(c) - \eta \text{BankValue}_t(c)],$$

subject to the fair-reserve override on scheduled exhaustive turns.

Verification then performs the only trusted state update:

$$B_{t+1} = \begin{cases} B_t \cup \{C(c_t)\}, & \forall (\mathfrak{E}, C(c_t), \pi_t) = 1, \\ B_t, & \text{otherwise.} \end{cases}$$

This pair of equations captures the architecture: sophisticated search chooses what to try, while verification chooses what becomes mathematics.

27 Correctness properties and limitations

27.1 Search/logic separation

Theorem 27.1 (Heuristic conservativity). *Assume the verifier is sound for the frozen environment. Altering Hilbert orientation, routing weights, Professor throttle, Dreamer policy, oracle hints, Depth Charge behavior, or fair/guided scheduling cannot by itself introduce a false theorem into BANK if every deposit still requires verifier acceptance.*

Proof. Every mathematical deposit is accepted by the same sound verifier. The listed mechanisms affect proposal order or resource allocation but not the verifier's acceptance relation. $\square$

27.2 No false claim of total decision

MAPQUEST is not a general terminating decision procedure for sufficiently expressive formal mathematics. If a proof exists and the fair assumptions hold, the system may eventually find it. Failure to find a proof under a finite budget establishes only that the run did not settle the target under that protocol.

27.3 No geometric miracle

Hilbert locality is useful only to the extent that the chosen representation places proof-relevant actions near one another. A poor embedding can make Hilbert traversal no better than an arbitrary permutation. Therefore any empirical performance claim must compare matched schedulers on the same legal candidate space and budget.

27.4 Settlement-distance computability

The exact r^* exists as a mathematical value on a finite-settling basin but need not be efficiently or totally computable on unrestricted theorem-search classes. The compass $\hat{r}$ is therefore an approximation whose calibration must be studied experimentally.

28 Implementation contract

A serious implementation should expose, at minimum, the following frozen interfaces:

Interface	Required contract
WFF encoder	Deterministic injective code $g(A)$ and canonical parser/pretty-printer round trip.
Rational address layer	Versioned map $a(A)$, least-preimage natural code $n(A)$, tuple pairing, collision tests.
Rule adapter	Declared arity, premise ordering, conclusion constructor, side-condition interface.
Witness enumerator	Effective enumeration of substitutions/unifiers/other instantiation evidence.
Hilbert engine	Deterministic $H_{k,p,o}$ and inverse/indexing behavior for each supported k, p, o .
Structural signature	Frozen theoremhood-independent feature computation and binning/scaling policy.
Compass	Versioned estimator $\hat{r}_t$ with no benchmark leakage.
Professor	Bounded score in $[0, 1]$, explicit throttle, no direct BANK write.
Dreamer	Sandboxed proposal generation with provenance and fixed budget.
Depth Charge	Trigger, budget, checkpoint/rollback semantics, verified-deposit exception.
Oracle layer	Exact intervention interface and provenance; disabled unless experimental arm declares it.
Sentinel	Deterministic or logged resource-policy decision with ALLOW/THROTTLE/VETO semantics.
Verifier	Small trusted checker for rule/certificate semantics, independent of heuristic scores.
Logger	Append-only event records sufficient to reproduce candidate ordering and accepted certificates.

29 Experimental program

The most important empirical question is not whether a Hilbert curve is aesthetically appealing. It is whether a meaningful representation plus Hilbert-local scheduling improves useful prefixes under fixed budgets.

A minimum ablation matrix should compare:

- lexicographic or database-order enumeration;
- random permutation with fixed seeds;
- breadth-controlled enumeration;
- Hilbert order on raw WFF rank;
- Hilbert order on structural addresses;
- Hilbert + settlement compass;
- Hilbert + compass + creativity;
- full T1P;
- T1P + Depth Charge;
- T1P + Dreamer;
- Professor throttle levels;
- selected oracle interventions as diagnostic upper bounds.

Hold fixed theorem split, environment hash, verifier, candidate legality rules, hardware, total compute, memory, stopping rules, and random seeds. Primary outcomes include verifier-valid settlement rate, expansions, wall time, certificate size, peak memory, duplicate rate, useful verified lemma yield, and interruption coverage.

30 Relation to the earlier framework

This specification consolidates several previously separate strands:

- Tenneson's explicit $\mathbb{N} \rightarrow \mathbb{Q}$ addressing and natural-number tuple coding;
- the locked convention that WFFs are zero-dimensional and rule arity supplies intrinsic inference geometry;
- the tagged coproduct of heterogeneous rule components;
- finite Hilbert scheduling of candidate premise tuples;
- the distinction between exact address/provenance and meaningful search metric;
- shortest-settlement Bellman control and approximate proof compasses;
- T1P, T2P, and Depth Charge;
- Creativity, Dreamer, the four oracle interfaces, Professor partial credit, Sentinel, BANK/FUTURE-BANK, and verifier sovereignty.

The new synthesis is the explicit division of labor:

Address :	What exact formal object is this?
Hilbert geometry :	Which inference neighborhoods are enumerated together?
Compass/Professor/T1P :	Which road should receive budget next?
Depth Charge/Dreamer :	How do we escape or rethink a bad route?
T2P/Sentinel :	What execution constraints must remain invariant?
Verifier :	Which proposed route steps are mathematically valid?

31 Conclusion

Hilbert–Tennessee MAPQUEST is not “move numerically toward the target address.” The target address is an exact destination identifier. The actual search domain consists of tagged candidate inference applications, each built from zero-dimensional formulas occupying ordered premise slots. Hilbert curves provide a systematic locality-preserving order over finite approximations to these rule-arity spaces. A witness fiber handles substitutions and side conditions. A target-conditioned settlement estimator, structural compass, Professor, creativity, and BANK value decide which regions deserve attention. T1P coordinates navigation; T2P freezes the experiment; Depth Charge provides bounded escape; Dreamer proposes counterfactual roads; oracle interfaces diagnose missing information; Sentinel controls resources; a fair channel preserves eventual access to finite proofs; and the verifier alone grants authority.

The resulting picture is precise enough to implement:

$\begin{aligned} &\text{WFF} \rightarrow \text{exact address} \rightarrow \text{tagged inference-address space} \rightarrow \text{Hilbert region} \\ &\quad \rightarrow \text{target-conditioned action} \rightarrow \text{verification} \rightarrow \text{BANK} \rightarrow \dots \rightarrow T. \end{aligned}$
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A successful proof is the resulting verifier-checkable route.

Source provenance and status

The architecture and terminology synthesized here are drawn from Brian Tennessee’s public research program, especially the manuscripts on explicit natural-to-rational addressing, Hilbert space-filling-curve theorem search, verified settlement search, shortest settlement/optimal control, reflective compass control, DATA MIND Dreamer and four-oracle synthesis, proof certificates, and the Tennessee protocols. The base/fiber formulation for schema witnesses and the present unified algorithmic packaging are graduate-level synthesis choices introduced here to close an implementation gap while preserving the project’s locked conventions. Claims labeled theorem are proved from their stated assumptions; claims about practical speedup remain experimental unless separately established.

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